

Nils Duran

AI & Machine Learning Engineer | MSc Data Science Student

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Summary

AI Engineer specializing in agentic workflows (LangGraph), LLM fine-tuning (LoRA), distributed Transformer training, and end-to-end product deployment. Proven experience improving Gemini models on clinical benchmarks and training deep networks on supercomputing infrastructure (MareNostrum 5). MSc Data Science student at UPC.

Experience

Sparsity Technologies , Data Engineering Intern	Barcelona, Catalonia Oct 2025 – Jan 2026 4 months
- Architected automated data collection and ingestion pipelines for high-performance graph databases. - Executed complex graph queries and data transformation workflows for competitor intelligence and market analysis. - Applied statistical analysis and NLP processing to optimize customer data alignment and SEO metrics.	
Werfen , Software Tester Intern (SW Quality & Automation)	Barcelona, Catalonia June 2025 – Sept 2025 4 months
- Engineered and maintained automated test suites in Java for in-vitro diagnostic (IVD) clinical software. - Authored unit, integration, and regression test suites ensuring compliance with medical quality regulations (ISO 13485). - Integrated automated test execution into CI/CD pipelines, accelerating release cycles and reducing regression time by 40%. - Managed defect lifecycle and triage in JIRA, performing root-cause analysis and edge-case boundary verification.	

Education

Mas- ter of Sci- ence Universitat Politècnica de Catalunya (UPC) , Data Science (MDS)	Barcelona, Catalonia Sept 2026 – present
Advanced specialization in distributed data systems, statistical learning, and machine learning infrastructure.	
Bachelor of Sci- ence Universitat Politècnica de Catalunya (UPC) , Artificial Intelligence	Barcelona, Catalonia Sept 2022 – June 2026
- Rigorous curriculum in Deep Learning, Reinforcement Learning, Algorithms, and Software Architecture. - Bachelor's Thesis (TFG): Computational neuroscience and EEG decoding with ST-GNNs and LoRA Transformers.	
Academic Exchange (Erasmus+) Karlsruhe Institute of Technology (KIT) , Informatics & Computer Science	Karlsruhe, Germany Oct 2024 – Mar 2025
Completed advanced coursework in high-performance algorithms and distributed systems in Germany. German C1 proficiency.	

Projects

Sinergia — Multi-Agent Systems & LLM Fine-Tuning	May 2025
- Applied AI collaboration project guided by Telefónica Innovación Digital. - Architected multi-expert conversational system with LangGraph & Gemini, boosting MedQA benchmark accuracy by +3% absolute (87% -> 90%). - Fine-tuned models with LoRA, computed agent diversity dissimilarity matrices, and built interactive Streamlit dashboard.	

Deep Learning on MareNostrum 5 Supercomputer

June 2024

- Distributed Vision Transformer training on the full MAMe dataset (>200 GB) using MareNostrum 5 infrastructure.
- Orchestrated multi-GPU training jobs via Slurm, optimized PyTorch data pipelines, achieving 0.75 F1-score across hundreds of classes.

Vegan Tools (vegantools.org)

Sept 2026

- Full-stack production web application integrating vision/NLP AI APIs (Product Checker, Menu OCR Interpreter, Recipe Veganizer).
- Engineered responsive UI, handled cloud hosting, custom domain DNS, and client-side state architecture.

IEEE BigData 2024 Cup — Chess Puzzle Difficulty

Nov 2024

- Ensemble ML model predicting human Elo difficulty of tactical chess puzzles leveraging Stockfish, Leela, and Maia engine heuristics.

Skills

AI Frameworks & LLMs: PyTorch, LangChain, LangGraph, LoRA / PEFT, Hugging Face Transformers, Gymnasium (RL), Scikit-learn

Engineering & MLOps: Python, Slurm (HPC), Docker, Git/GitHub, Vector Databases (Pinecone), REST APIs, Streamlit, Linux

Data & Backend: PostgreSQL, SQL, Apache Spark, Graph Databases (Neo4j), NumPy, Pandas

Languages: Catalan (Native), Spanish (Bilingual), English (C2 Proficient), German (C1 Advanced)